# MATH 401. MIDTERM #1. Sadullaev L.

- I promise that I will not use any external resources while working on this exam. I will not search the internet for any hints on the problems in the exam, and I will not look at any textbook, notes, handouts, or use online search and online resources. I will not also communicate with any one else about this exam while working on the same.

- As promised, I did not use any external resources while working on this exam.

00:20 am 10/11/2023.

## Problem 1.

Prove the following statements for sets $A, B, C$, and for the family of sets $\mathcal{A}$.

$$A \Delta B = (A \setminus B) \cup (B \setminus A)$$

(1) $$(A \Delta B) \cap C = (A \cap C) \Delta (B \cap C)$$

(2) $$\left( \bigcup_{A_i \in \mathcal{A}} A_i \right)^c = \bigcap_{A_i \in \mathcal{A}} A_i^c$$

**Solution:**

Using subset including going both ways:

(1) $(\subseteq)$

Let arbitrary $x \in (A \Delta B) \cap C \Rightarrow$

$x \in A \Delta B$ and $x \in C \Rightarrow$ (def. of $\cap$)

$\Rightarrow x \in (A \setminus B) \cup (B \setminus A)$, $x \in C \Rightarrow$ (def. of $\Delta$)

$\Rightarrow x \in A \setminus B$ or $x \in B \setminus A$, $x \in C \Rightarrow$ (def. of $\cup$)

(4) $\Rightarrow x \in A, x \notin B$ or $x \in B, x \notin A$ and $x \in C$ (def of $\setminus$)

from here important point:

If $x \in A$, $x \notin B$ and $x \in C \Rightarrow x \in (A \cap C) \setminus (B \cap C)$

If $x \notin A$, $x \in B$ and $x \in C \Rightarrow x \in (B \cap C) \setminus (A \cap C)$

(def.s of $\cap$ & $\setminus$)

$\Rightarrow$

Either

$x \in (A \cap C) \setminus (B \cap C)$ or $x \in (B \cap C) \setminus (A \cap C) \Rightarrow$

$\Rightarrow x \in (A \cap C) \setminus (B \cap C) \cup (B \cap C) \setminus (A \cap C)$ (def of $\cup$)

(9) $\Rightarrow x \in (A \cap C) \Delta (B \cap C)$ (def. of $\Delta$)

(10) $\Rightarrow (A \Delta B) \cap C \subseteq (A \cap C) \Delta (B \cap C)$

(2)

Let arbitrary $x \in (A \cap C) \Delta (B \cap C)$. $\Rightarrow$

$x \in (A \cap C) \setminus (B \cap C) \cup (B \cap C) \setminus (A \cap C)$ (def. of $\Delta$)

$\Rightarrow x \in (A \cap C) \setminus (B \cap C)$ or $x \in (B \cap C) \setminus (A \cap C)$ (def of $\cup$)

$\Rightarrow x \in A \cap C, x \notin B \cap C$ or $x \in B \cap C, x \notin A \cap C$ (def of $\setminus$)

$\Rightarrow$ (13) $\left[\begin{array}{l} x \in A,\ x \in C,\ x \notin B \text{ or } \boxed{x \notin C} \\ x \in B,\ x \in C,\ x \notin A \text{ or } \boxed{x \notin C} \end{array}\right.$ or (def. of $\cap$)

(not possible, since $x \in B$, $x \in C$ has been set in the beginning)

(14) $\Rightarrow \left[\begin{array}{l} x \in A,\ x \in C,\ x \notin B \\ x \in B,\ x \in C,\ x \in A \end{array}\right.$ or $\Rightarrow$

(15) $\left[\begin{array}{l} x \in A,\ x \in C,\ x \notin B \Rightarrow x \in A \setminus B,\ x \in C \\ x \in B,\ x \in C,\ x \notin A \Rightarrow x \in B \setminus A,\ x \in C \end{array}\right.$ or $\Rightarrow$

$\Rightarrow x \in A \setminus B$ or $x \in B \setminus A$ , $x \in C$ $\Rightarrow$

$x \in (A \setminus B) \cup (B \setminus A)$ , $x \in C$ (def. of $\cup$)

$\Rightarrow x \in (A \Delta B)$, $x \in C$ (def of $\Delta$)

$\Rightarrow x \in (A \Delta B) \cap C$ (def. of $\cap$)

(16)
$\Rightarrow (A \Delta B) \cap C \supseteq (A \cap C) \Delta (B \cap C)$

(10) & (16) $\Rightarrow$

(1) $(A \Delta B) \cap C = (A \cap C) \Delta (B \cap C)$

$\boxtimes$

(2) similar technique, subset inclusion:

($\subseteq$)

(17) Let arbitrary $x \in \left( \bigcup_{A_i \in \mathcal{A}} A_i \right)^c$ $\Rightarrow$

(18) $x \notin \bigcup_{A_i \in \mathcal{A}} A_i$ $\Rightarrow$ (def. of compl)

$\Rightarrow x \notin A_i$ <u>for all</u> $A_i \in \mathcal{A}$ (def. of not being in $\cup$)

$\Rightarrow x \in A_i^c$ for all $A_i \in \mathcal{A}$ (def. of complem.)

$\Rightarrow x \in \bigcap_{A_i \in \mathcal{A}} A_i^c$ (def. of $\cap$)

(20) $\Rightarrow \left( \bigcup_{A_i \in \mathcal{A}} A_i \right)^c \subseteq \bigcap_{A_i \in \mathcal{A}} A_i^c$

(2)

(21) Let arbitrary $x \in \bigcap_{A_i \in \mathcal{A}} A_i^c$. $\Rightarrow$

$\Rightarrow x \in A_i^c$ for all $A_i \in \mathcal{A}$ (def. of $\cap$)

$\Rightarrow x \notin A_i$ for all $A_i \in \mathcal{A}$ (def. of com.)

$\Rightarrow x \notin \bigcup_{A_i \in \mathcal{A}} A_i$ (def. of $\notin \cup$)

$\Rightarrow x \in \left( \bigcup_{A_i \in \mathcal{A}} A_i \right)^c$ (def. of comp.)

(22) $\Rightarrow \left( \bigcup_{A_i \in \mathcal{A}} A_i \right)^c \supseteq \bigcap_{A_i \in \mathcal{A}} A_i^c$

(20) & (22) gives what we wanted. $\blacksquare$.

**Problem 2.**

Consider the relation $R$ defined as follows:

(1) $$R = \{(x,y) \in \mathbb{R} \times \mathbb{R} \mid x^2 + y^2 = 1\}$$

Check if $R$ is an equivalence relation.

**Solution:**

Equivalence relation defined on set $X$ is the one satisfying the following 3 properties:

(i) Reflexivity : $x \sim x, \quad \forall x \in X$

(ii) Symmetry : $x \sim y \Rightarrow y \sim x, \quad \forall x, y \in X.$

(iii) Transitivity : $x \sim y, y \sim z \Rightarrow x \sim z, \quad \forall x, y, z \in X.$

Let's check set $R$ through these 3 tests:

(i) Let $r = (x, y) \in R \Rightarrow$

$$x^2 + y^2 = 1 = x^2 + y^2 \Rightarrow r \sim r \quad \checkmark$$

(ii) Let $r = (x_r, y_r), q = (x_q, y_q) \in R$ s.t

$r \sim q \Rightarrow x_r^2 + y_r^2 = 1 = x_q^2 + y_q^2$ (flipping the sides:)

$$\Rightarrow x_q^2 + y_q^2 = 1 = x_r^2 + y_r^2 \Rightarrow q \sim r \quad \checkmark$$

(iii) Let $r = (x_r, y_r), p = (x_p, y_p), q = (x_q, y_q) \in R$ s.t.

$r \sim p, \quad p \sim q.$ That is

$$r \sim p \Rightarrow \underline{x_r^2 + y_r^2 = 1} = x_p^2 + y_p^2,$$

$$p \sim q \Rightarrow x_p^2 + y_p^2 = \underline{1 = x_q^2 + y_q^2}$$

By combining the underlined parts:

$x_r^2 + y_r^2 = 1 = x_q^2 + y_q^2 \Rightarrow r \sim q \checkmark$. All three holds!

Hence $R$ is an equivalence relation

**Problem 3.**

Let $S$ be the set of all points on a straight line in $\mathbb{R}^2$. Is the set $S$ countable? Justify!

**Solution:**

Let $S = \{(x, y) \in \mathbb{R}^2 \mid ax + by = c$ for fixed $a, b, c \in \mathbb{R}\}$

First easy approach I would offer is NO. $S$ is not countable, since $S$ is a set of all points from the cartesian product of 2 uncountable sets, $\mathbb{R} \times \mathbb{R}$. We learnt in LSIRA 1.6, I can't remember exactly the proposition number (Geuss 1.6.2) "Cartesian product of 2 countable sets is countable", and saw $\mathbb{N} \times \mathbb{Z}$ was countable. My argument is contrapositive of this above mentioned proposition. Since $\mathbb{R}$ & $\mathbb{R}$ are uncountable, so is their cartesian product, making the set $S$ uncountable.

If you think it is not sufficient enough, because I'm using the result directly from the text, I would offer the proof technique we used for $(0,1) \subset \mathbb{R}$ to be uncountable.

- To be easier, let me cut $S$ to take its subset $S'$ of all reals lying on the line $S' = (0,1) \times (0,1)$. If we show $S'$ is uncountable, it follows that $S' \subset S$ its superset $S$ must be uncountable too.

For contradiction, assume $S'$ is **countable**. $\Rightarrow$

$\exists$ a listing for $S'$ to order **all its elements**. (def. of countability)

So,

(1) $S' = \{(x,y) \in (0,1)\times(0,1) \mid ax+by=c$ for fixed $a,b,c \in \mathbb{R}\}$

Let that ordered-fashion $S'$ listing be

(2) $S' = \{(x_1,y_1),(x_2,y_2),(x_3,y_3),\dots \mid ax+by=c$ for some $a,b,c \in \mathbb{R}\}$.

Since all $x_i$'s & $y_i$'s $\in \mathbb{R}$ for convinience let me write down $x_i$'s & $y_i$'s in the following 2 columns:

(3)

| $x$ | | $y$ | |
|---|---|---|---|
| $0.x_{11}\,x_{12}\,x_{13}\,x_{14}\dots$ | ; | $0.y_{11}\,y_{12}\,y_{13}\,y_{14}\dots$ | $(x_1,y_1)$ |
| $0.x_{21}\,x_{22}\,x_{23}\,x_{24}\dots$ | ; | $0.y_{21}\,y_{22}\,y_{23}\,y_{24}\dots$ | $(x_2,y_2)$ |
| $0.x_{31}\,x_{32}\,x_{33}\,x_{34}\dots$ | ; | $0.y_{31}\,y_{32}\,y_{33}\,y_{34}\dots$ | $(x_3,y_3)$ |
| $\dots\dots\dots$ | ; | $\dots\dots\dots$ | $\dots$ |
| $0.x_{i1}\,x_{i2}\,x_{i3}\,x_{i4}\dots$ | ; | $0.y_{i1}\,y_{i2}\,y_{i3}\,y_{i4}\dots$ | $(x_i,y_i)$ |

Now we can create a different pair of $x,y$'s from all the pair of elements (elt.s) in the listing above.

Let $x = 0.a_1\,a_2\,a_3\dots$, $y = 0.b_1\,b_2\,b_3\dots$

To find the decimals of $a_i$'s of $x$:

(4) $$a_i = \begin{cases} 1, & \text{if } x_{ii} = 2 \\ 2, & \text{if } x_{ii} \neq 2 \end{cases} \qquad \forall i \in \mathbb{N}.$$

Similarly,

for the decimal $b_i$'s of $y$:

$$(5)\ b_i = \begin{cases} 3, & \text{if } y_{ii} = 2 \\ 4, & \text{if } y_{ii} \neq 2 \end{cases}$$

This new poin $(x, y) = (0.a_1 a_2 a_3 \ldots, 0.b_1 b_2 b_3 \ldots)$ differs at least by the i-th decimal in their respective listing points $(x_i, y_i)$. So we could find a new point $(x, y)$ when we assumed we included **all the points of $S'$** in the listing (2) $\Rightarrow$ Contradiction.

That means $S'$ is not countable, so is $S$.

**Problem 4.**

(a) Consider the sequence $x_n = \{\frac{1}{n}\}$ over the set $X = (0,1] \cap \mathbb{Q}$. Show that the sequence $\{x_n\}$ is Cauchy.
Is the sequence convergent?

(b) Let $\{a_n\}$ be a Cauchy sequence in $\mathbb{R}$. Show that this sequence has a subsequence $\{a_{n_k}\}$ s.t.

$$|a_{n_{k+1}} - a_{n_k}| < \frac{1}{2^k} \qquad \forall k \in \mathbb{N}.$$

**Solution:**

(a)

(1) Let $\{x_n\} = \{\frac{1}{n}\} \in X = (0,1] \cap \mathbb{Q}$.

WTS: $\{x_n\}$ is Cauchy

WTS: $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall n, k \geq N$

(2) $|x_n - x_k| < \varepsilon.$

First let's see the difference of 2 successive terms:

(3) $|x_k - x_{k-1}| = \left|\frac{1}{k} - \frac{1}{k-1}\right| = \left|\frac{-1}{k(k-1)}\right| = \frac{1}{k(k-1)}$

Then, (2) has the form as following, if we consider $k \geq n$

$$|x_n - x_k| = |x_k - x_n| = |x_k - x_{k-1} + x_{k-1} - x_{k-2} + x_{k-2} + \dots +$$

$$+ x_{n+1} - x_n| \overset{(\Delta \text{ ineq-y})}{\leq} |x_k - x_{k-1}| + |x_{k-1} + x_{k-2}| + \dots + |x_{n+1} - x_n|$$

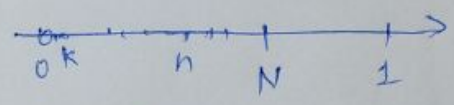
$$= \overset{\nearrow 0}{\frac{1}{k(k-1)}} + \overset{\nearrow 0}{\frac{1}{(k-1)(k-2)}} + \dots + \overset{\nearrow 0}{\frac{1}{(n+1)n}} = 0 + \dots + 0 = 0$$

for sufficiently large enough $k, n \geq N$. $\Rightarrow$

$\{x_n\}$ is Cauchy. $\boxtimes$

Is $\{x_n\}$ Convergent. – YES.

Let $x_k = 0$ when $k \to \infty$ $x_k \to 1/\infty \to 0$ although it never reaches. Then




$$|x_n - x_k| = |x_n - 0| = |x_n| = \frac{1}{n} = \varepsilon \quad N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1$$

(If we let $x_k = 0$ & $n$ very close to $N$ ($n = N+1$) we would be considering the possibly the biggest difference $|x_n - x_k|$ $\forall k, n \in N$)

$\Rightarrow \{x_n\}$ is convergent, but the converging point $0 \notin X$.

(b) Let $\{a_n\}$ be Cauchy.

WTS: $\{a_n\}$ has a subsequence $\{a_{n_k}\}$ s.t.

$$|a_{n_{k+1}} - a_{n_k}| < \frac{1}{2^k}$$

$\{a_n\}$ is Cauchy $\Rightarrow$

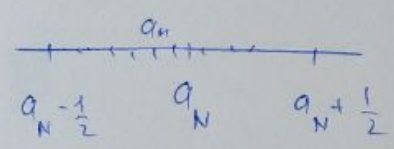
$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $\forall n, k \geq N$

(1) $|a_n - a_k| < \varepsilon$

Now let $k = N$ & $\varepsilon = \frac{1}{2}$ then (1) is

(2) $|a_n - a_N| < \frac{1}{2}$ $\forall n \geq N$ $\Rightarrow$

(3) $a_N - \frac{1}{2} < a_n < a_N + \frac{1}{2}$ $\forall n \geq N$.




So, all $a_n$'s $\in \left[a_N - \frac{1}{2}, a_N + \frac{1}{2}\right]$

I'm going to employ similar proof technique in BW theorem from here.

Let $I_1 = \left[a_N - \frac{1}{2}, a_N + \frac{1}{2}\right]$, (pay attention the length of $I_1 = 1$, I'll denote it as $|I_1| = 1$)

Pick first element here as $a_{n_1} \in I_1$. Then let $I_2 = \frac{I_1}{2}$ s.t contains infinitely many terms of $I_1$, first half or second half. Let $a_{n_2}$ be the first elt. of $\{a_n\}$ after $a_{n_1}$ in $I_2$, and divide $I_2$ from the middle again. let $I_3 = \frac{I_2}{2}$ which contains infinitely many elts of $\{a_n\}$ & let $a_{n_3}$ be the first elt. of $\{a_n\}$ after $a_{n_2}$ in $I_3$, and so on.

If we pay attention $|a_{n_{k+1}} - a_{n_k}| \leq \frac{|I_1|}{2^k}$ (at most) which we know $|I_1| = 1$. $\Rightarrow$

$$|a_{n_{k+1}} - a_{n_k}| < \frac{1}{2^k} \qquad \forall\, k \in \mathbb{N}.$$

$\blacksquare$

**Problem 5.**

Use the definition of continuity $(\varepsilon, \delta > 0)$ of a function directly to show that if $f, g, h \to \mathbb{R}$ are all continuous functions at $x = a$, then the function $p(x) = f \cdot g + h$ is also continuous at $x = a$.

**Solution:**

Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be continuous functions at $x = a$.

WTS: $p(x) = f(x) \cdot g(x) + h(x)$ is also cont. at $x = a$.

That is $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.

(1) $|p(x) - p(a)| < \varepsilon$ whenever $|x - a| < \delta$.

(2) $|p(x) - p(a)| = |f(x) \cdot g(x) + h(x) - (f(a) \cdot g(a) + h(a))| =$

$= |[f(x) \cdot g(x) - f(a) \cdot g(a)] + [h(x) - h(a)]| \leq$ (using $\Delta$ ineq.y)

$\leq |f(x) \cdot g(x) - f(a) g(a)| + |h(x) - h(a)| =$

$= |[f(x) g(x) - f(x) \cdot g(a)] + [f(x) \cdot g(a) - f(a) \cdot g(a)]| + |h(x) - h(a)| \leq$ ($\Delta$ inequality)

$\leq |f(x)| \cdot |g(x) - g(a)| + |g(a)| \cdot |f(x) - f(a)| + |h(x) - h(a)|$

Then (2) comes to form:

(3) $|p(x) - p(a)| = |f(x) \cdot g(x) + h(x) - f(a) \cdot g(a) - h(a)| \leq$ (after 2 $\Delta$ inequality) $+ |h(x) - h(a)|$

$\leq |f(x)| \cdot |g(x) - g(a)| + |g(a)| \cdot |f(x) - f(a)| + |h(x) - h(a)|$

For (3) here we may have 4 possible cases:

(a) $f(a) = 0$, $g(a) = 0$

(b) $f(a) = 0$, $g(a) \neq 0$

(c) $f(a) \neq 0$, $g(a) = 0$

(d) $f(a) \neq 0$, $g(a) \neq 0$

(a) If $f(a)=0$, $g(a)=0$, then (3) is

(3) $|p(x)-p(a)| \le |f(x)|\cdot|g(x)-\overset{\nearrow 0}{g(a)}| + |\overset{\nearrow 0}{g(a)}|\cdot|f(x)-f(a)| + |h(x)-h(a)| \Rightarrow$

$|p(x)-p(a)| \le |f(x)|\cdot|g(x)-0| + 0\cdot|f(x)-0| + |h(x)-h(a)| \Rightarrow$

(4) $|p(x)-p(a)| \le |f(x)|\cdot|g(x)| + |h(x)-h(a)|$

(5) As $f(x)$ is con-s at $x=a$ $\exists\, \delta_f>0$ s.t. $|f(x)-\overset{=0}{f(a)}| = |f(x)| < \sqrt{\frac{2\varepsilon}{3}}$

(6) As $g(x)$ is con-s at $x=a$ $\exists\, \delta_g>0$ s.t. $|g(x)-\overset{=0}{g(a)}| = |g(x)| < \sqrt{\frac{2\varepsilon}{3}}$

(7) As $h(x)$ is con-s at $x=a$, $\exists\, \delta_h>0$ s.t $|h(x)-h(a)| < \frac{\varepsilon}{3}$

Then, by letting $\delta = \min\{\delta_f, \delta_g, \delta_h\}$, (4) is:

$|p(x)-p(a)| \le |f(x)|\cdot|g(x)| + |h(x)-h(a)| < \sqrt{\frac{2\varepsilon}{3}}\cdot\sqrt{\frac{2\varepsilon}{3}} + \frac{\varepsilon}{3} = \varepsilon.$

(8) $|p(x)-p(a)| < \varepsilon \Rightarrow$ $p(x)$ is con-s at $x=a$.

(b) If $f(a)=0$, $g(a)\neq 0$, then (3) is

(3) $|p(x)-p(a)| \le |f(x)|\cdot|g(x)-\overset{=0}{g(a)}| + |g(a)|\cdot|f(x)-\overset{\nearrow 0}{f(a)}| + |h(x)-h(a)| \Rightarrow$

(9) $|p(x)-p(a)| \le |f(x)|\cdot|g(x)-g(a)| + |g(a)|\cdot|f(x)| + |h(x)-h(a)|$

(10) As $f(x)$ is con-s at $x=a$, $\exists\, \delta_f>0$ s.t. $|f(x)-\overset{=0}{f(a)}| = |f(x)| < \frac{\varepsilon}{3\cdot|g(a)|}$

(11) As $g(x)$ is con-s at $x=a$, $\exists\, \delta_g>0$ s.t. $|g(x)-g(a)| < |g(a)|$.

(12) As $h(x)$ is con-s at $x=a$, $\exists\, \delta_h>0$ s.t. $|h(x)-h(a)| < \frac{\varepsilon}{3}$.

Then, by letting $\delta = \min\{\delta_f, \delta_g, \delta_h\}$, (9) is:

$|p(x)-p(a)| < \frac{\varepsilon}{3\cdot|g(a)|}\cdot|g(a)| + |g(a)|\cdot\frac{\varepsilon}{3|g(a)|} + \frac{\varepsilon}{3} = \frac{\varepsilon}{3}+\frac{\varepsilon}{3}+\frac{\varepsilon}{3} = \varepsilon$

(15) $|p(x)-p(a)| < \varepsilon \Rightarrow p(x)$ is con-s at $x=a$.

(c) If $f(a) \neq 0$, $g(a)=0$, then (3) is

(3) $|p(x)-p(a)| \leq |f(x)| \cdot \underbrace{|g(x)-g(a)|}_{\to 0} + \overbrace{|g(a)|}^{\to 0} \cdot |f(x)-f(a)| + |h(x)-h(a)|$

$\Rightarrow$

(16) $|p(x)-p(a)| \leq |f(x)| \cdot |g(x)| + |h(x)-h(a)|$

Similar case to case (a), (4) look previous page $\Rightarrow$

(17) $|p(x)-p(a)| < \varepsilon \Rightarrow p(x)$ is con-s at $x=a$.

(d) If $f(a) \neq 0$, $g(a) \neq 0$, then (3) is

(3) $|p(x)-p(a)| \leq |f(x)| \cdot |g(x)-g(a)| + |g(a)| \cdot |f(x)-f(a)| + |h(x)-h(a)|$

As $f(x)$ is con-s at $x=a$, $\exists\, \delta_{1f} > 0$ s.t. $|f(x)| < 2|f(a)|$

As $g(x)$ is con-s at $x=a$, $\exists\, \delta_g > 0$ s.t. $|g(x)-g(a)| < \dfrac{\varepsilon}{3 \cdot 2|f(a)|}$

As $f(x)$ is con-s at $x=a$, $\exists\, \delta_{2f} > 0$ s.t. $|f(x)-f(a)| < \dfrac{\varepsilon}{3|g(a)|}$

As $h(x)$ is con-s at $x=a$, $\exists\, \delta_h > 0$ s.t. $|h(x)-h(a)| < \dfrac{\varepsilon}{3}$.

Then by letting $\delta = \min\{\delta_{1f}, \delta_g, \delta_{2f}, \delta_h\}$ (3) is:

$$|p(x)-p(a)| \leq |f(x)| \cdot |g(x)-g(a)| + |g(a)| \cdot |f(x)-f(a)| + |h(x)-h(a)| <$$

$$< \cancel{2 \cdot |f(a)|} \cdot \frac{\varepsilon}{3 \cdot \cancel{2|f(a)|}} + \cancel{|g(a)|} \cdot \frac{\varepsilon}{3\cancel{|g(a)|}} + \frac{\varepsilon}{3} = \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \Rightarrow$$

$\Rightarrow$

(18) $|p(x)-p(a)| < \varepsilon \Rightarrow p(x)$ is con-s at $x=a$.

By combining all results (8), (15), (17), (18) for all 4 possible cases (a), (b), (c), (d) respectively, $p(x)$ is cont-s at $x=a$.

## Problem 6.

Assume that the sequence $\{a_n\}$ is nonnegative and converges to $a$. Also, assume that for the sequence $\{b_n\}$, $\limsup_{n\to\infty} b_n = b < \infty$ is positive. Show that $\limsup_{n\to\infty} a_n b_n = ab$.

**Solution:**

Assume $\{a_n\}$ is nonnegative $\Rightarrow$

(1) $a_n \geq 0 \quad \forall n \in \mathbb{N}$

(2) $\{a_n\} \to a \quad (\text{by } (1),\ a \geq 0)$

Also assume

(3) $\limsup_{n\to\infty} b_n = b < \infty, \quad b > 0.$

WTS: $\limsup a_n b_n = ab \iff \forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t $\forall n \geq N$ $|a_n b_n - ab| < \varepsilon$

(4) $\{a_n\} \to a \iff \begin{cases} \limsup_{n\to\infty} a_n = a \\ \liminf_{n\to\infty} a_n = a \end{cases} \Rightarrow \limsup_{n\to\infty} a_n = a$

Then an easy approach, if you consider as true would be:

(5) $\limsup_{n\to\infty} a_n b_n = \underbrace{\limsup_{n\to\infty} a_n}_{a} \cdot \underbrace{\limsup_{n\to\infty} b_n}_{b} = a \cdot b.$ $\blacksquare$

(But, I was worried I haven't used their $(a_n, b_n)$ being positive) $\Rightarrow$

So, I might offer second approach:

By (4), we have

(6) $\limsup a_n = a \quad (a \geq 0)$

Let

(7) $A_n = \sup\{a_k \mid k \geq n\} \quad \forall n \in \mathbb{N}$ &

(8) $B_n = \sup\{b_k \mid k \geq n\} \quad \forall n \in \mathbb{N}$. Then

(9) $\limsup\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} A_n = a, \; a \geq 0$ (def. of limsup),

(10) $\limsup\limits_{n\to\infty} b_n = \lim\limits_{n\to\infty} B_n = b < \infty \; (b > 0)$.

From them it follows that.

(11) $\limsup\limits_{n\to\infty} a_n \cdot b_n = \lim\limits_{n\to\infty} A_n B_n$

If $A_n = 0 \; \forall n \in \mathbb{N} \Rightarrow a_n = 0 \; \forall n \in \mathbb{N} \Rightarrow a = 0$.

Then by WTS, $\forall \varepsilon > 0, \; \exists N \in \mathbb{N}$ s.t.

(12) $|a_n b_n - ab| = |\overset{\to 0}{A_n} \overset{\to 0}{B_n} - \overset{0}{a}\overset{0}{b}| = |0 - 0| = 0 < \varepsilon, \; N = 1$

(very trivial case, we can set $N = 1$).

If $A_n \neq 0$ even for large enough $n$, then

by aiy WTS part

$\forall \varepsilon > 0, \ \exists N \in \mathbb{N}$ s.t

(13) $|a_n b_n - ab| = |A_n B_n - ab| = |(A_n \cdot B_n - A_n \cdot b) + (A_n \cdot b - ab)| \leq$

$\leq |A_n(B_n - b)| + |b(A_n - a)| =$ (triangle $\Delta$ ineq.)

$= |A_n||B_n - b| + |b||A_n - a|$ .

So

(14) $|a_n b_n - ab| \leq |A_n||B_n - b| + |b| \ |A_n - a|$

But then, by (10), $\lim_{n\to\infty} B_n = b \Rightarrow$

$\forall \frac{\varepsilon}{2|A_n|} > 0, \ \exists N_1 \in \mathbb{N}$ s.t. $\forall n \geq N_1$

(15) $|B_n - b| < \frac{\varepsilon}{2|A_n|}$ (pay attention, we are considering $A_n \neq 0$ case.)

And by (9), $\lim_{n\to\infty} A_n = a$ $(a \geq 0)$

$\forall \frac{\varepsilon}{2|b|} > 0, \ \exists N_2 \in \mathbb{N}$ s.t. $\forall n \geq N_2$

$|A_n - a| < \frac{\varepsilon}{2|b|}$ (pay attention by (3), $b \neq 0$)

Then, continuing (14) by letting $N = \max\{N_1, N_2\}$ above,

$|a_n \cdot b_n - ab| \leq |A_n| \ |B_n - b| + |b||A_n - a| < |A_n| \cdot \frac{\varepsilon}{2|A_n|} + |b| \cdot \frac{\varepsilon}{2|b|} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$

$\Rightarrow |a_n \cdot b_n - ab| < \varepsilon \Rightarrow \lim_{n\to\infty} \sup a_n b_n = \lim_{n\to\infty} A_n B_n = ab.$ ∎